

Project Phase 1

Behavioural Wakefulness Verification System (BWVS)

An Intelligent Non-Contact Wakefulness Verification System using 24 GHz mmWave Radar

Boys Batch -2

Team Members

1. Aswin Raaj .R (23BEE030)
2. Mani Balaji .S (23BEE043)
3. Tharun .V.S (23BEE057)

Guided by

Dr.G.Pandiaraj

Assistant Professor (Sl. Grade)

Electrical and Electronics Engineering

Problem Statement

The Problem:

Conventional alarm systems assume that waking up is a single event. However, in reality:

- ☐ Users often stop the alarm and immediately return to sleep.
- ☐ Existing alarms only verify alarm dismissal.
- ☐ Temporary body movement is mistaken as genuine wakefulness.
- ☐ False wake-ups reduce productivity and affect daily routines.

Core Problem

- ☐ Existing alarm systems cannot reliably distinguish between temporary awakening and sustained wakefulness.

Motivation

Why is this important?

Daily issues caused by ineffective wake-up systems:

- ☐ Missing classes or work
- ☐ Sleep inertia after alarm dismissal
- ☐ Repeated snoozing
- ☐ Lack of behavioural verification
- ☐ No monitoring after alarm completion

Current systems answer:

"Was the alarm stopped?"

Our project answers:

"Is the user actually awake?"

Objective

Primary Objective :

Develop an embedded system capable of verifying sustained wakefulness using non-contact sensing.

Specific Objectives :

- ☐ Detect human presence without wearable devices
- ☐ Differentiate temporary movement from purposeful activity
- ☐ Introduce a cognitive wake-up challenge
- ☐ Monitor wakefulness after alarm dismissal
- ☐ Reduce false wake-up confirmations
- ☐ Log wakefulness sessions for later analysis

Literature Survey

Existing Method	Limitation
Conventional Alarm Clock	Stops after button press
Smartphone Alarm	Easily snoozed
PIR Motion Alarm	Detects only large movement
Camera-based Monitoring	Privacy concerns
Wearable Devices	User dependent
Smart Home Motion Detection	Cannot verify sustained wakefulness

Research Gap

No affordable embedded system verifies sustained behavioural wakefulness using non-contact sensing.

PIR vs mmWave

Feature	PIR	HLK-LD2410C mmWave
Detect Stationary Human	✗	✓
Works in Darkness	✓	✓
Motion Sensitivity	Low	High
Detection Reliability	Moderate	High
Micro-motion Detection	✗	✓
Distance Tracking	✗	✓
Suitable for Wakefulness Verification	Limited	Excellent

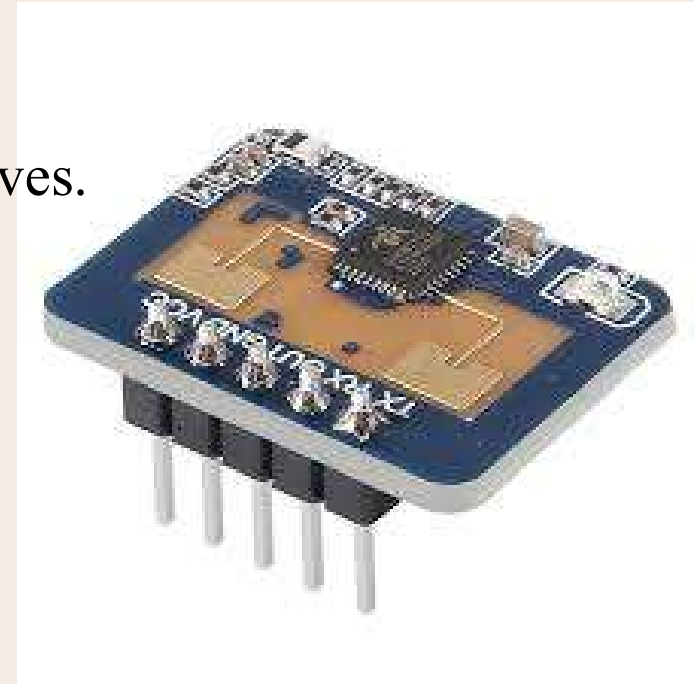
Selected Sensor: HLK-LD2410C

Why mmWave Radar?

The HLK-LD2410C radar continuously emits 24 GHz frequency-modulated electromagnetic waves. By analyzing reflected signals, it can detect:

- ☐ Human presence
- ☐ Distance
- ☐ Macro movement
- ☐ Micro movement
- ☐ Stationary things

Unlike PIR, it does not depend on infrared radiation.



The radar divides space into multiple configurable distance gates, allowing continuous tracking of human presence and movement.

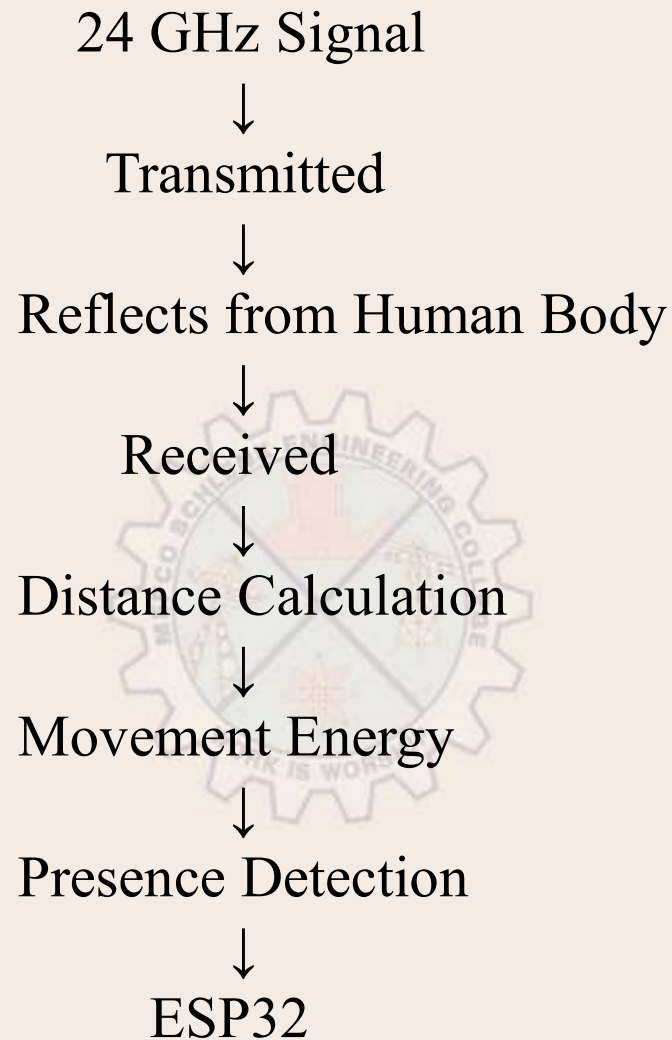
mmWave Radar Data Sheet Study – HiLink(HLK) LD2410C

- ☐ Detection angle: +/-60 deg
- ☐ Resolution: 0.75m each (0 to 20 gates) – configured separately through HLK app via Bluetooth
- ☐ Detection Range: 6m
- ☐ Operating Voltage: 3.3v , Current Consumption: 80mA
- ☐ Pins: TX, RX, GND, VCC, OUT (0 – human present, 1 – absent)

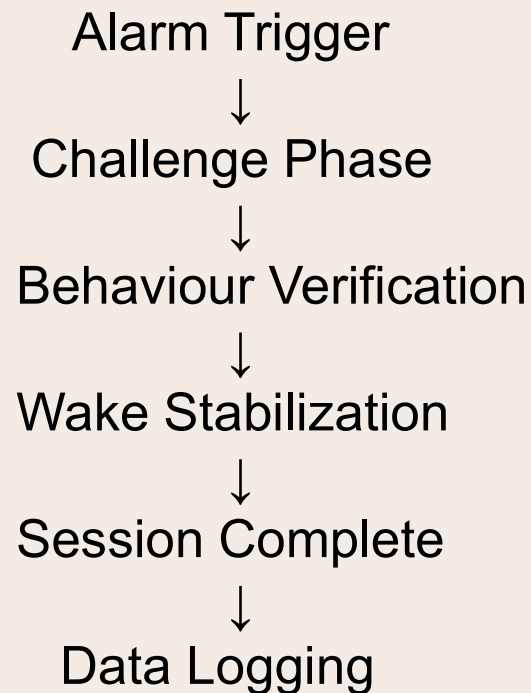
UART Communication: - sends a packet containing,

- ☐ Target State (0-no target, 1-moving target, 2-stationary target, 3-moving+stationary)
- ☐ Movement target distance
- ☐ Stationary distance
- ☐ Movement energy
- ☐ Stationary energy
- ☐ Detection distance

Working Principle of mmWave



Proposed Methodology



Challenge Phase

- ☐ Activates cognitive response

Verification Phase

- ☐ Observes behavioural activity

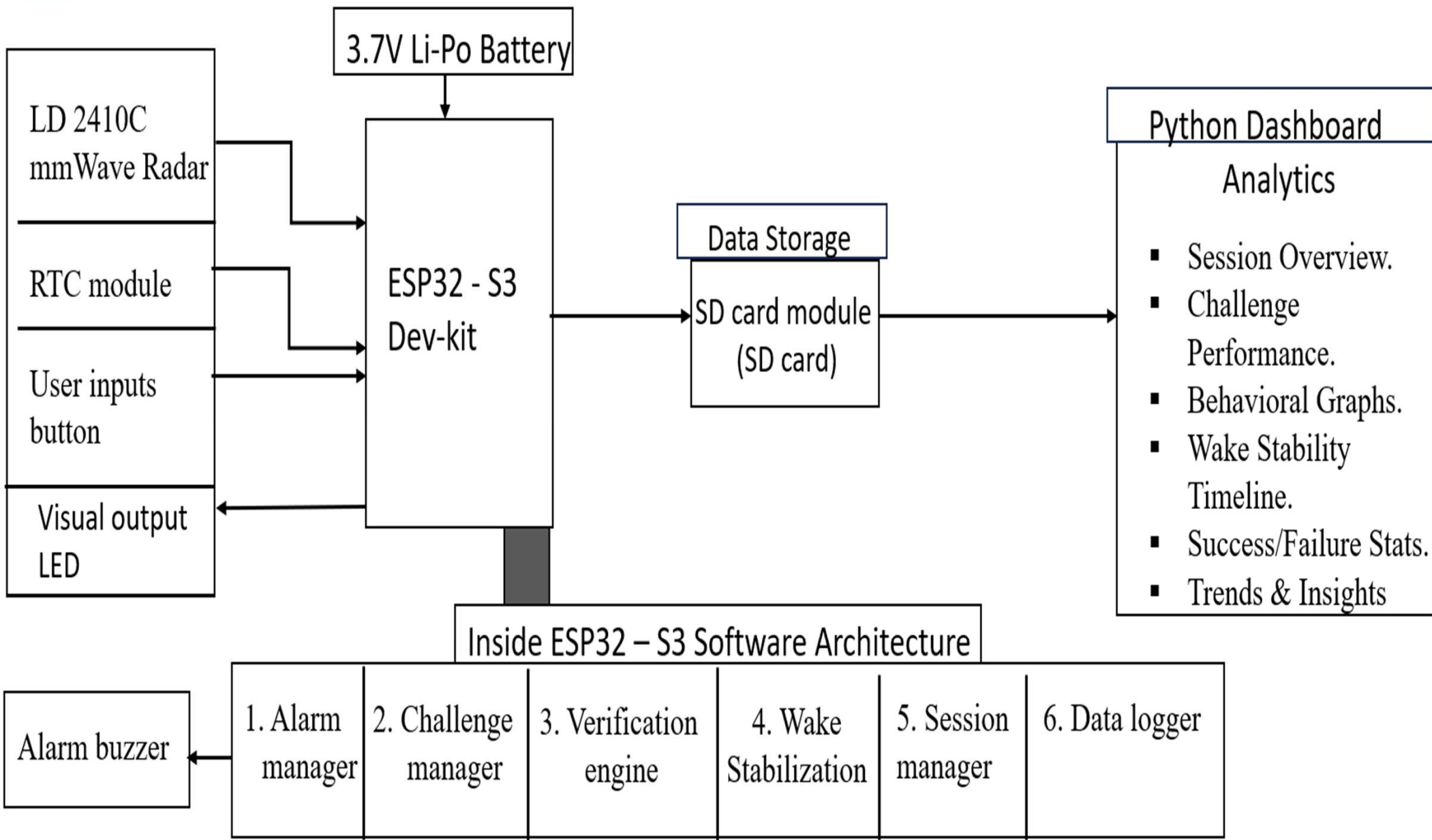
Stabilization Phase

- ☐ Detects possible relapse into sleep

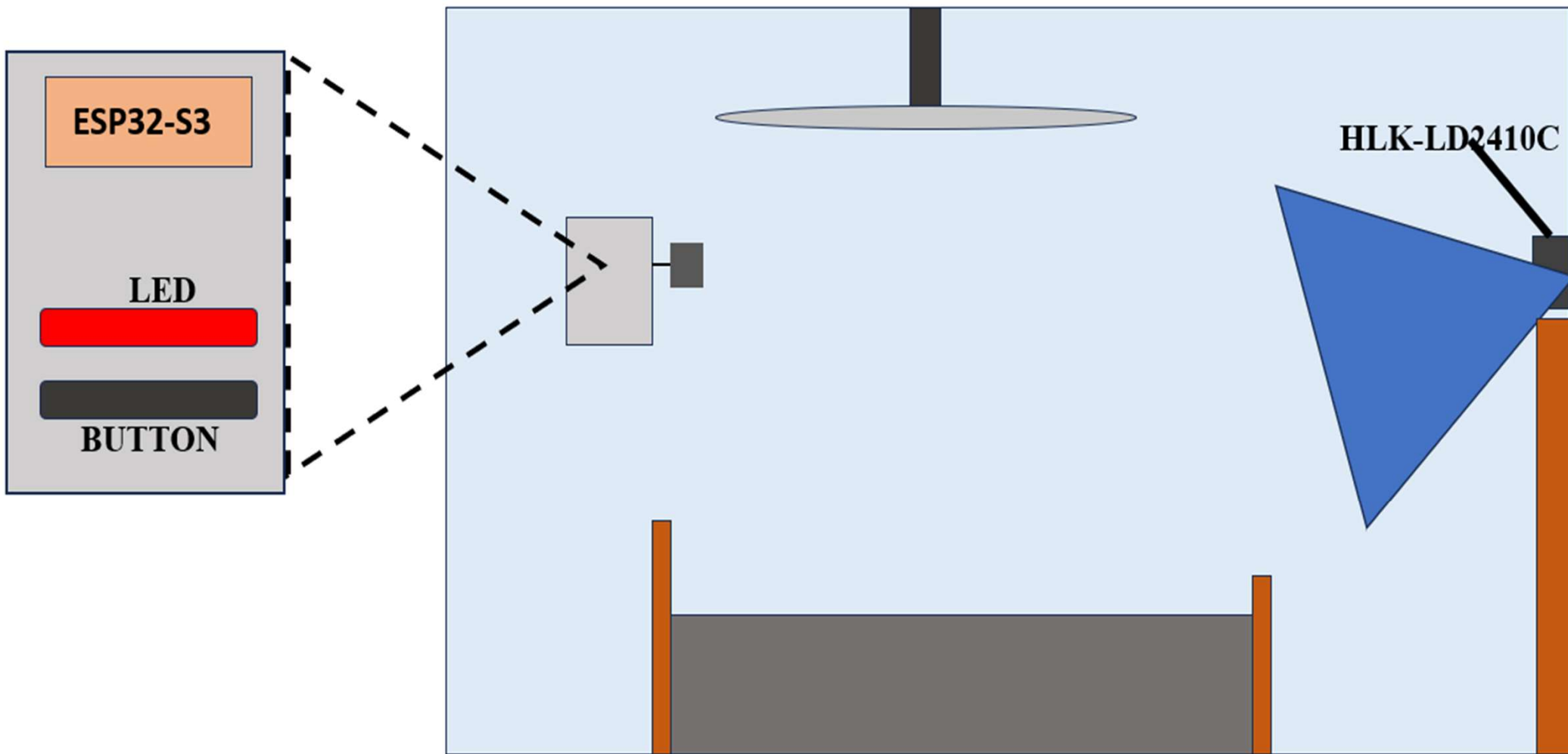
Logging

- ☐ Stores wakefulness session for analysis

Block Diagram



System Design



Hardware components

Component	Purpose
ESP32 – S3	Main controller
HLK-LD2410C	Presence & movement sensing
RTC Module	Alarm scheduling
LEDs	Challenge indication
Push Buttons	Cognitive challenge
Buzzer	Alarm output
LDR	Ambient light detection (future support)
SD Card Module	Session logging
Power Supply	System power

Advantages

- ☐ Non-contact operation
- ☐ Privacy preserving
- ☐ Detects stationary humans
- ☐ Detects micro movements
- ☐ Behavioural verification instead of simple motion detection
- ☐ Multi-stage wake-up approach
- ☐ Data logging support

Future Enhancements

- ☐ Behavioural confidence scoring
- ☐ Adaptive wakefulness models
- ☐ Machine learning based personalization
- ☐ Multi-sensor data fusion
- ☐ Smart room integration
- ☐ Long-term wakefulness analytics



GANTT CHART

Sl. No.	Task	July-26	Aug-26	Sep-26	Oct-26	Nov-26
1.	Title Selection, System Design, Components Selection					
2.	Components Purchase and Assemble, System firmware design					
3.	Testing and Validating prototype against edge cases					
4.	Final Prototype and Report Submission					

References

Hao, Z., Wang, Y., Li, F. *et al.* Detection of vital signs based on millimeter wave radar. *Sci Rep* **15**, 28112 (2025). <https://doi.org/10.1038/s41598-025-09112-w>

En-Kang Wu, Qi-Gao Fan, Ming-Chao Li, Ji-Hao Zhang, Jie Jia, Tian Qiang, Cong Wang, Xiao-Feng Gu, Jun-Ge Liang,
Non-contact monitoring of human cardiorespiratory activity during sleep using FMCW millimeter wave radar, 2025, 116144,
ISSN 0263-2241, <https://doi.org/10.1016/j.measurement.2024.116144>.

M. Alizadeh, G. Shaker, J. C. M.c D. Almeida, P. P. Morita and S. Safavi-Naeini, "*Remote Monitoring of Human Vital Signs Using mm-Wave FMCW Radar*," in *IEEE Access*, vol. 7, pp. 54958-54968, 2019, doi: 10.1109/ACCESS.2019.2912956.

Thank You



28 July 2026

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